**NAMANGA SECONDARY SCHOOL**

**FORM TWO MOCK EXAMINATION AUGUST 2020**

**033** **BIOLOGY**

**TIME: 2:30 HRS** **Wednesday 5th August, 2020 am.**

**INSTRUCTIONS:**

1. This paper consist of section A, B and C
2. Answer all questions in section A & B and **one** question from section C
3. All answer must be written in the spaces provided
4. All writings must be in blue or black ink **EXCEPT** for the diagrams which must be in pencil

## SECTION A (30 MARKS)

1. Select the correct answer in each item given and write its letter in the box provided
   i. The structure of the leaf that allows oxygen and carbon dioxide to diffuse into and out of the leaf during gaseous exchange is known as
      a) Mesophyll b) Cuticle c) Epidermis d) Stomata
   ii. In mammals, deoxygenated blood from the body enter the heart through
      a) Right auricle b) Left auricle c) Right ventricle d) Left ventricle
   iii. The green Plants in a community are Ecologically referred to as …….
      a) Consumers b) Decomposer c) Producer d) Ecosystem
   iv. A kidney is an example of ……
      a). A tissue b). A system c). An organ d) An organelle
   v. The feeding relationship represented by the sequence of organism:
      "Green plants → Grasshopper → Frog" the arrows show the ….
      a) Flow of en energy b) Flow of nutrient c) Flow of ecosystem
      d) Flow of minerals
   vi. Which of the following is a risk behavior?
      a). Alcoholism b). Being patient c). Shaking hands with HIV infected individuals
      d) Sharing toilet facilities with HIV infected individuals
   vii. A food sample Y turned blue – black when it was mixed with iodine solution. This conclude that the sample Y contained:
      a).Fats and Oil b).Proteins c). Reducing sugar d) Starch
   viii. Which of the following is expected to happen if blood supply to the brain is reduced
      a). Fainting b). Nausea c). Shock d) Vomiting
   ix. The part of a light microscope used for controlling amount of light is the
      a). Base b). Diaphragm c). Eyepiece d) Mirror

   x. In human being, the brain is an example of
      a). Tissue b). An organ c). Organelles d) System

| i | Ii | Iii | Iv | V | vi | vii | viii | Ix | x |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

2. For each of the items (i) – (x), write T**RUE** if a statement is correct or **FALSE** if the statement is not correct in a space provided.

i. Superior vena cava carries deoxygenated blood from the upper part of the body to the heart……

ii. Gaseous exchange in mammals takes place in the nose and mouth ……

iii. One enzyme can act on several food substances ……

iv. Kingdom is the lowest rank of classification……

v. Anaemia is a condition in which the patient has few erythrocytes ……
vi. Saprophytes are organisms which feed on dead organic matter example Plasmodium……
vii. Unicellular organisms are made up of one cell only……
viii. A consumer in ecosystem produces their own food ……
ix. Syphilis is communicable disease ……
x. All bacteria are harmful to man ……

3. For each of items i – v, choose the correct answer among the given alternative in list B, and write its letter in the table below.

| LIST A | LIST B |
|---|---|
| i. A disease transmitted through sexual intercourse and treated by antibiotics<br>.<br>ii. A viral disease transmitted through sexual intercourse<br><br>iii. An disease characterized by fever and coughing up blood sputum<br><br>iv. disease transmitted by contaminated water<br><br>v. A disease caused by lack of proper hygiene and victim had a stool of rice – water like coloration. | A. Cholera<br>B. Typhoid<br>C. AIDs<br>D. Arteries<br>E. Syphilis<br>F. Common cold<br>G. Schist soma<br>H. Tuberculosis<br>I. Veins<br>J. Malaria<br>K. Vena cava |

| **A** | i. | ii. | iii. | iv. | v. |
|---|---|---|---|---|---|
| **B** | | | | | |

4. Complete each of the following statement by writing the correct answer in the answer Sheet provided
   i. All arteries transport oxygenated blood except…….
   ii. The blood vessel which carries deoxygenated blood from different parts of the body to the heart is called…….
   iii. …… is the red pigment in erythrocytes which have high affinity with oxygen
   iv. …… is a type of blood cancer caused by over production of white blood cells
   v. ……. is an extended epidermal cells in plants which absorb water and mineral salts from soil

**SECTION B (50 MARKS)**

Short answer questions

5. (a) What is:
   i. Food web
   ii. Food chain
   (b) Below is the list of organisms occupying a certain habitat in an ecosystem
   Grass, Lion, Zebra and Gazelle
   i. Construct a food web using the given organisms above
   ii. Mention one group of organism which must be added to balance the ecosystem
   (c). Name organism(s) which are;
   i. Tertiary consumer……
   ii. Primary producers ……..
6. (a) What is respiration?
   (b) Name and define two types of respiration
   (c) Write four factors affecting the rate of respiration
   (d) Outline four areas/places where anaerobic respiration could be applied in our daily real life situation
7. Figure 01 represent a digestive system of a mammal study it carefully then answer the questions that follow

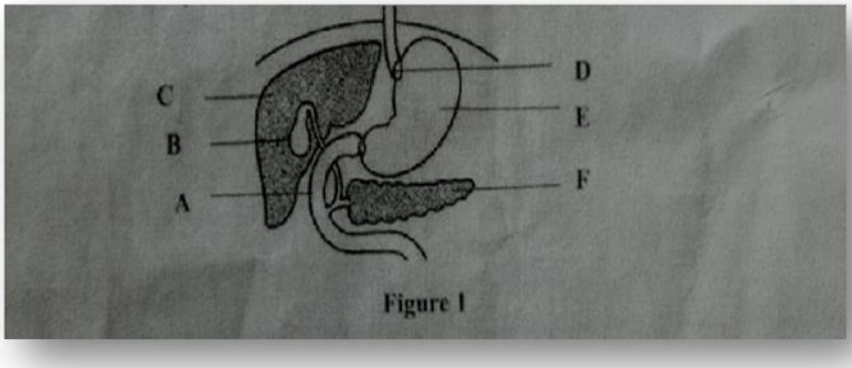


Figure 1

(a) Name the parts labeled with letter A, B, C, D, E and F.
(b) Name two enzymes secreted by the part labeled with letter E and F.
(c) Give four difference between animal cell and plant cell

8. (a) State the meaning of the term "tissue "as used in biology.
(b) Arrange the following term in a sequential order from simple to complexity of cell Differentiation. Tissue, organelle, organism, organ, cell and system.
(c). Draw the diagram of animal cell and on it;
i. Label the part which controls all activity of the cell
ii. Label the part which protects the structures inside the cell

9. (a). With one example outline phyla of kingdom fungi
(b). Outline any four (4) advantages of kingdom fungi

**SECTION C (20 MARKS)**
(Essay questions)
Answer only **one** question in this section

10. Write an essay on HIV/AIDs by using the following guidelines
(a) Meaning of HIV/AIDs.
(b) Causes of AIDs.
(c) Mode of transmission.
(d) Signs and symptoms of AIDs.
(e) Prevention and control of AIDs.
(f) Effects of HIV/AIDs

11. Write an essay about photosynthesis under the following guidelines
i. Site of the process
ii. Raw materials
iii. Conditions
iv. End products
v. Importance of this process to other organisms